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Superseding
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MILITARY SPECIFICATION

REELING MACHINE, CABLE, MOTOR DRIVEN, RL 172( )/G

1. SCOPE

1.1 Scope.- This specification covers one type of lightweight reel unit for picking up and paying out field wire on Reel RL-159( )/U, designated as Reeling Machine, Cable Motor Driven, RL-172( )/G. (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-B-571	Bolt, nuts, studs, and tap-rivets"and materials for same".
QQ-S-781	Steel strapping, flat.
TT-A-580	Anti-seize compound, white lead base, general purpose (for threaded fittings).
UU-P-271	Paper, wrapping, waterproofed kraft.
PPP-B-621	Boxes, wood, nailed and lock corner.
PPP-C-843	Cushioning material,cellulosic.
PPP-P-291	Paperboard, wrapping, cushioning.
PPP-T-60	Tape, pressure sensitive adhesive, waterproof for packaging and sealing.
PPP-T-97	Tape, pressure sensitive adhesive, filament reinforced

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MIL-P-116	Preservation, methods of.
MIL-B-117	Bags, interior, packaging.
MIL-E-5272	Environmental testing, Aeronautical and Associated Equipment, General Specification for.
MIL-M-13231	Marking of electronic items.
MIL-F-14072	Finished for ground signal equipment.
MIL-E-55301	Electromagnetic compatibility.

STANDARDS

MILITARY

MIL-STD-105	Sampling procedures and tables for inspection by attributes.
MIL-STD-129	Marking for shipment and storage.
MIL-STD-169	Extreme-temperature cycle.
MIL-STD-170	Moisture resistance test cycle for ground signal equipment.
MIL-STD-731	Quality of wood members for containers and pallets.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-425300	Reeling Machine, Cable, Motor Driven RL-172( )/G.

(Copies of specifications, standards, and drawings required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Reeling Machine, Cable, Motor Driven RL-172( )/G is a single axle reel unit powered by a 24 volt dc motor which can be operated from military trucks having a 24 volt dc electrical system. It shall be capable of laying wire at all speeds up to 35 mph and picking up wire at all speeds up to 15 mph, dependent on the condition of the terrain.

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3.2 Construction.- The Reeling Machine, Cable, Motor Driven, RL-172( )/G shall be constructed in accordance with SC-DL-425300.

3.3 Finish, protective.- The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.3).

3.4 Marking.- Marking shall conform to MIL-M-13231. Each reeling machine shall be serial numbered. (See 4.3).

3.5 Cleaning.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.6 Fastenings.- Swaging, peening, or staking (prick punching) of threads or other parts, except as specifically required by the applicable drawings or specifications, or the use of "Locktite" on threads will not be permitted.

3.7 Anti-Seize compound.- Anti-seize compound conforming to TT-A-580 shall be used to prevent seizing of threads in all cases where the threads surface is aluminum.

3.8 Operation requirements.

3.8.1 Mounting.- The equipment shall payout and pick up wire when mounted as specified in 4.8.1.

3.8.2 Capacity.- The equipment shall have the capacity to handle one Reel RL-159( )/U containing one mile of Wire WD-1/TT. (See 4.8.2).

3.8.3 Power source.- The equipment shall be capable of meeting the requirements of 3.8.4 thru 3.8.11 when operated from any military vehicle having a 24 volt dc electrical system. (See 4.8.3).

3.8.4 Variable speed control.- The operating level shall control the tension applied to the slack "V" belt and shall smoothly vary the speed of the reel axle when in the "tension on" position by means of a spring applied braking force. In the "tension off" position, the axle shall be free from drag. The lever shall be held in the "on" or "off" position by a spring loaded toggle action. (See 4.8.4).

3.8.5 Tension lever.- The tension lever shall provide a drag on the reel axle when in the "tension on" position by means of a spring applied braking force. In the "tension off" position, the axle shall be free from drag. The lever shall be held in the "on" or "off" position by a spring loaded toggle action. (See 4.8.5).

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3.8.6 Brake lever.- A braking force shall be applied to the reel axle when the brake lever is released and with the tension lever in the operating "tension on" position. It shall apply a sufficient force to stop the axle when rotating 300 rpm. (See 4.8.6).

3.8.6.1 Regardless of the position of the "tension lever", the suddenly applied force specified in 4.8.6.1 shall stop the rotation of the loaded axle from a speed of 300 rpm to 0 rpm within 3 revolutions.

3.8.7 Hand crank.- The operating lever shall be capable of being removed from its normal position on the equipment and attached to the reel axle for use as a hand crank for manual operation of the equipment. The hand crank shall be capable of providing sufficient leverage to rotate the axle manually when the axle has a fully loaded Reel RL-159( )/U mounted thereon and with the tension lever in the "tension off" position. (See 4.8.7).

3.8.8 Overload.- The slack "V" belt, specified in 3.8.4, shall slip when the feed reel is stopped. There shall be no damage to the equipment or overloading of the motor. (See 4.8.8).

3.8.9 Payout of wire.- When paying out wire, the tension lever shall be in the "tension on" position and the motor shall not be running. The equipment shall function so that the operator can prevent over-run of the reel at all speeds. (See 4.8.9).

3.8.9.1 When paying out the wire, the tension lever shall be in the "tension on" position and the motor shall not be running. The equipment shall function so that the operator can prevent over-run of the reel at all speeds. (See 4.8.9.1).

3.8.10 Pickup of wire.- The equipment, with the tension lever in the "tension off" position shall pick up Wire WD-1/TT by running into the wire without overrunning or snagging of the wire. The wire shall be carried over the roof of the vehicle to the reeling machine by means of the front and rear mast assemblies, using hand pad to level wind wire on the RL-159( )/U Reel. (See 4.8.10).

3.8.10.1 The one mile of wire specified in 4.8.10.1 shall be dragged straight in by the equipment by the power of the motor, using hand pad, level wind wire on the Reel RL-159( )/U. If snagging of the wire occurs, the equipment reel shaft may stall but the fuses shall not blow. (See 4.8.10.1).

3.8.11 Continuous operation.- The equipment shall withstand continuous operation for a period of 24 hours, at room temperature, without failure. (See 4.8.11).

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3.9 Service conditions.- The equipment shall meet the following service conditions:

3.9.1 Temperature.- The equipment shall meet the requirements of 3.8.4 thru 3.8.8 at steps 4 and 8 of MIL-STD-169. After step 10, the equipment shall meet the requirements of 3.8.4 thru 3.8.8. There shall be no evidence of deterioration. (See 4.9.1).

3.9.2 Moisture resistance.- (See 4.9.2). During and after the test specified in 4.9.2, the equipment shall meet the requirements of 3.8.4 thru 3.8.8. The equipment shall show no evidence of deterioration.

3.9.3 Rain.- (See 4.9.3). During and after the test specified in 4.9.3, the equipment shall meet the requirements of 3.8.4 thru 3.8.8. The equipment shall show no evidence of deterioration.

3.9.4 Vibration.- (See 4.9.4). With the equipment shock mounts (if any) blocked, the amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. After vibration the equipment shall be examined for defects covered in 4.6 and meet the specified requirements of 3.8.4 thru 3.8.8.

3.9.5 Bounce.- (See 4.9.5). The equipment shall meet the specified requirements of 3.8.4 thru 3.8.8 after subjection to the test specified in 4.9.5. No physical damage except minor surface abrasion shall result from this test.

3.9.6 Shock.- The equipment shall meet the specified requirements of 3.8.4 thru 3.8.8 with no physical damage after subjection to the test specified in 4.9.6.

3.9.7 Dust.- After subjection to the test specified in 4.9.7, the equipment shall be examined for defects covered in 4.6 and shall meet the specified requirements of 3.8.4 thru 3.8.8. There shall be no deterioration of the equipment.

3.10 Suppression of radio interference.- The equipment motor shall be capable of meeting the radiated and conducted requirements of MIL-E-55301 for tactical support equipment. The motor manufacturer is required to supply certified test data and a schematic showing the location, type and value of each component used to meet the above requirements. (See 4.10).

3.11 Canvas cover.- The canvas cover shall fit properly before and after being subjected to the test specified in paragraph 4.5.

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3.12 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, characteristic, etc. (See 4.7).

3.13 Technical literature, tools and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order.

3.14 Government-loaned property.- The following equipment shall be Government-loaned to perform the tests specified in 4.8 and 4.9. (Wire and Splicing Sleeves to be expendable).

(a) Six Reels RL-159( )/U each filled with one mile of Wire WD-1/TT.

(b) Any one of the following vehicles:

1/4 ton vehicle
3/4 ton vehicle
2-1/2 ton vehicle

(c) One each Splicing Tool, TL-582( )/G, with 3 boxes of Splicing Sleeves.

(d) One Reel Unit RL-31-E.

3.15 Workmanship.- The item shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs: (See 4.6).

3.2 Construction
3.5 Cleaning
3.6 Fastenings
3.7 Anti-Seize compound

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Inspection covered by subsidiary documents. (See 4.3).

(b) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.4).

(2) Quality conformance inspection of preparation for delivery. (See 4.11).

4.3 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.3
Marking	3.4

4.4 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.3 and 4.4.1 thru 4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. (See 6.5). Test equipment for Government verification inspection shall be made available by the contractor.

4.4.1 Group A inspection.- This inspection, including sampling, shall conform to Table I and the inspection procedures of MIL-STD-105 using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government.

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Table I - Group A inspection

Inspection	Rqt. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.15	4.6	1.0%	4.0%
Operational				
Capacity	3.8.2	4.8.2)		
Power source	3.8.3	4.8.3)	1.0%	
Variable speed control	3.8.4	4.8.4)	for	
Tension lever	3.8.5	4.8.5)	the	
Brake lever	3.8.6	4.8.6)	group	
Hand crank	3.8.7	4.8.7)		
Overload	3.8.8	4.8.8)		

4.4.2 Group B inspection.- This inspection, including sampling, shall conform to Table II and to the special procedures for small-sampling inspection of MIL-STD-105. The AQL shall be 6.5% defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

Table II - Group B inspection

Inspection	Rqt. Para.	Insp. Para.
Interchangeability	3.12	4.7

4.4.3 Group C inspection.- This inspection shall conform to Tables III and IV, and shall be performed on sample units that have been subjected to and met group A and group B inspection. Sample units shall be selected at random without regard to their quality except that the samples selected at the start of the contract shall be selected from the first units produced.

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4.4.3.1 Group C-1 inspection.- This inspection shall consist of the tests specified in Table III and shall be performed on 2 units every 3 months or every 600 units produced whichever occurs first.

Table III - Group C-1 Inspection

Inspection	Rqt Para.	Insp. Para.
Road tests		
Mounting	3.8.1	4.8.1
Payout of wire	3.8.9, 3.8.9.1	4.8.9, 4.8.9.1
Pickup of wire	3.8.10, 3.8.10.1	4.8.10, 4.8.10.1
Continuous operation	3.8.11	4.8.11
Canvas covers	3.11	4.5
Shock	3.9.6	4.9.6
Suppression of radio interference	3.10	4.10

4.4.3.2 Group C-2 inspection.- This inspection shall consist of the tests specified in Table IV and shall be performed on 2 units every year or every 3000 units, whichever occurs first, except the Moisture resistance test 4.9.2 which shall be performed one time only at the beginning of production. The 2 units shall be subjected to all of the tests listed and in the order stated below:

Table IV - Group C-2 inspection

Inspection	Rqt. Para.	Insp. Para.
Extreme temperature	3.9.1	4.9.1
Moisture resistance	3.9.2	4.9.2
Rain	3.9.3	4.9.3
Vibration	3.9.4	4.9.4
Bounce	3.9.5	4.9.5
Dust	3.9.7	4.9.7

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4.4.3.3 Noncompliance.- The contractor shall immediately report, in writing, each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.4.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.5 Canvas cover.- The canvas cover shall be inspected for satisfactory fit both before and after the following test: Soak the cover in tap water at normal room temperature for 24 hours, and then dry thoroughly at normal room temperature. (See 3.11).

4.6 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Table V. (See 3.15).

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Table V - Classification of visual and mechanical defects

Classification	Defects
Major	Welds not of sufficient penetration. Rough edges on metal. Cover fasteners missing or inoperable. Cover fit too tight. Broken stitches on cover. Holes in cover fabric.
Minor	Loose nut and screws. Loose rivets. Scratches, cuts, abrasions, etc. causeing exposure of bare metal. Color of cover not as specified. Marking not as specified. Stains on cover fabric.

4.7 Inspection for dimensional interchangeability.- The equipment shall be examined to determine compliance with the physical interchangeability requirements of 3.12.

4.8 Operational tests.-

4.8.1 Mounting.- The equipment shall be capable of being mounted either vertically on the tailgate or horizontally on the floor of the 1/4 ton, the 3/4 ton and the 2-1/2 ton vehicles. While in either position the equipment shall meet the requirements of 3.8.1.

4.8.2 Capacity.- A Reel RL-159( )/U, loaded with one mile of Wire WD-1/TT, shall be placed in operating position on the equipment to determine compliance with the requirements of 3.8.2.

4.8.3 Power source.- The reeling machine,with a Reel RL-159( )/U loaded with one mile of Wire WD-1/TT, shall be tested to determine compliance with the requirements of 3.8.3.

4.8.4 Variable speed control.- With a Reel RL-159( )/U loaded with one mile of Wire WD-1/TT placed in operating position on the reeling machine, the rpm of the axle, with motor operating from the 24 volt d.c.source, shall be checked using a standard laboratory tachometer. The operating lever shall be operated to determine compliance with the requirement specified in 3.8.4.

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4.8.5 Tension lever.- With a Reel RL-159( )/U loaded with one mile of Wire WD-1/TT, placed in operating position on the reeling machine, the tension lever shall be tested to determine compliance with the requirements specified in 3.8.5.

4.8.6 Brake lever.- When the axle is turning at 300 rpm loaded with one mile of Wire WD-1/TT on Reel RL-159( )/U and the "tension on", release the speed control lever to determine compliance with the requirement specified in 3.8.6. This test shall be performed with the axle turning both in the clockwise and counterclockwise direction.

4.8.6.1 With one mile of Wire WD-1/TT on Reel RL-159( )/U and the axle turning at 300 rpm, as stated above, a suddenly applied force to the hand brake shall be applied to determine compliance with 3.8.6.1.

4.8.7 Hand crank.- The operating lever shall be removed, attached to the axle and turned manually for 10 revolutions in both directions to determine compliance with the requirement specified in 3.8.7.

4.8.8 Overload.- One mile of Wire WD-1/TT wound on Reel RL-159( )/U shall be mounted in Reel Unit RL-31-E. The wire shall be fed to the reeling machine. During the pickup, the reel, feeding the wire to the equipment, shall be machanically stopped a minimum of three times to determine compliance with the requirement specified in 3.8.8.

4.8.9 Payout of wire.- With the equipment mounted as specified in 4.8.1, and with the vehicle traveling a winding course over rough terrain at speeds of 15mph to 35 mph, one mile of Wire WD-1/TT shall be paid out. The payout of the wire shall be controlled utilizing the brake control. (See 3.8.9).

4.8.9.1 One mile of wire shall be paid out in a straight-away over fields or plowed land with the equipment mounted as specified in 4.8.1 at speeds of 15 mph to 35 mph and meet the requirements of 3.8.9.1. The payout of the wire shall be controlled utilizing the brake control.

4.8.10 Pickup of wire.- After performing the test specified in 4.8.9, the one mile of wire shall be picked up with the equipment mounted as specified in 4.8.1, and running into the wire at speeds of 5 mph to 15 mph using the masts, front and rear, to determine compliance with the requirement specified in 3.8.10.

4.8.10.1 After performing the test of 4.8.9.1, the one mile of wire shall be dragged in by the reeling machine without manual assistance.. The vehicle shall be stationary and its back to the end of the wire. (See 3.8.10.1).

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4.8.11 Continuous operation.- The equipment loaded with one mile of Wire WD-1/TT on Reel RL-159( )/U shall be operated continously for a period of 24 hours and meet the requirements specified in 3.8.11.

4.9 Service tests.-

4.9.1 Temperature test.- The equipment shall be subjected to the temperature cycle shown on MIL-STD-169. At Step 4 and Step 8, the tests specified in paragraphs 4.8.4 thru 4.8.8 shall be conducted. After Step 10, the tests specified in 4.8.4 thru 4.8.8 shall be conducted. The equipment shall meet the requirements of 3.9.1

4.9.2 Moisture resistance test.- The equipment shall be tested as follows:

Test method.-

(a) Dry at 130°F± 5° for 24 hours.

(b) Condition at 77° F ± 5° and 40 to 50 percent relative humidity for 24 hours.

(c) Conduct tests in accordance with 4.8.4 thru 4.8.8.

(d) Subject to continuous cycling for fifteen 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. Repeat tests of (c) above every five (5) days.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° F± 5° and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. Conduct tests per 4.8.4 thru 4.8.8.

4.9.3 Rain test.- The equipment shall run continuously throughout the following test.

4.9.3.1 Place the equipment, in its normal operating position, in a test chamber maintained at 77° F ± 5°. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° F ± 5° and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

4.9.3.2 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for 1 hour.

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4.9.3.3 Rotate the equipment 90 degrees horizontally to expose a second side of the equipment to the wind and rain. Maintain this condition for 1 hour.

4.9.3.4 Continue as in 4.9.3.3 until all four sides of the equipment have been exposed to the wind and rain, each side for 1 hour.

4.9.3.5 During and immediately after completion of the foregoing, the equipment shall be tested per paragraphs 4.8.4 thru 4.8.8.

4.9.4 Vibration test.- The equipment shall be tested for resonant frequencies from 10 to 55 cycles per second to determine compliance with 3.9.4 as follows:

(a) The equipment shall be fastened in its vertical mounting position simulating the mounting specified in 4.8.1 on a vibration table that can be controlled within 10 percent of the specified amplitude. The vibration table shall provide approximately sinusoidal vibration.

(b) The equipment shall be vibrated successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) Upon completion of vibration, inspect visually and mechanically per 4.6. Then conduct tests per 4.8.4 thru 4.8.8.

4.9.5 Bounce test.- The equipment shall be secured in its vertical mounting position simulating the mounting specified in 4.8.1 to a vehicular adapter plate and placed on the table of the Package Tester, as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The plate shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm $\pm$ 1 percent, for a total of 3 hours. The adapter plate shall be rotated throught 90 degrees at the end of each 3/4 hour period, in the same direction. The equipment shall be inspected visually and then tested in accordance with paragraphs 4.8.4 through 4.8.8. Upon completion of the above test, the equipment shall meet the requirements of 3.9.5.

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4.9.6 Shock.- With the equipment in its horizontal position, raise one end to an angle of approximately 45 degrees. Drop the unit on to a 4 x 4 inch fir timber placed crosswise to catch the raised end. The timber shall be backed by concrete or a rigid steel frame. Repeat the drop three times on each end. At the completion of the above test, the equipment shall meet the requirements specified in 3.9.6.

4.9.7 Dust.- The equipment shall be subjected to Sand and Dust Test, Procedure 1, Part 1 in accordance with MIL-E-5272. The equipment shall be operated at 300 + 0 - 50 rpm for the duration of this test. Upon completion of test, the equipment shall be stopped and started without adjustment and tested in accordance with paragraphs 4.8.4 thru 4.8.8 and visual inspection per 4.6 to determine compliance with 3.9.7.

4.10 Suppression of radio interference.- The equipment motor shall be tested in accordance with the radiated and conducted test of MIL-E-55301 for tactical support equipment and meet the requirements of 3.10.

4.11 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

4.12 Rough handling test (preparation for delivery).- When rough handling test, in accordance with MIL-P-116, is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

4.8.2 Capacity
4.8.3 Power source.
4.8.4 Variable speed control
4.8.5 Tension lever
4.8.6 Brake lever
4.8.7 Hand crank

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified in the contract or order. (See 6.2).

5.1.1 Level A.

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5.1.1.1 Cleaning.- Each reeling machine shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Each reeling machine shall be dried in accordance with MIL-P-116.

5.1.1.3 Unit packaging.- Unit protection shall be in accordance with MIL-P-116 and as specified herein.

5.1.1.3.1. Technical manuals.- Each technical manual shall be packaged in accordance with item 1 of figure 1 and Table VI.

5.1.1.3.2 Cable, cable guide, Pad LC-80, V belt, and wrench shall be packaged Method III in accordance with Table VI.

5.1.4 Level C.- Component parts and technical manuals shall be packaged in a manner which will afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier's commercial practice will be acceptable when it fulfills the above requirements.

5.2 Packing.- Packing shall be level A, B, or C as specified in the contract or order. (See 6.2).

5.2.1 Level A.- Each reeling machine shall be packed in accordance with items 2 through 7 of figure 1 and Table VI. Technical manuals and component parts packaged as specified in 5.1.1.3.1 and 5.1.1.3.2 shall be packed within the container.

5.2.2 Level B.- Same as 5.2.1 except that the box shall be class 1.

5.2.3 Level C.- Component parts and technical manuals packaged as specified in 5.1 and reeling machine shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers shall be in accordance with the rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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REELING MACHINE RL-172( )/G

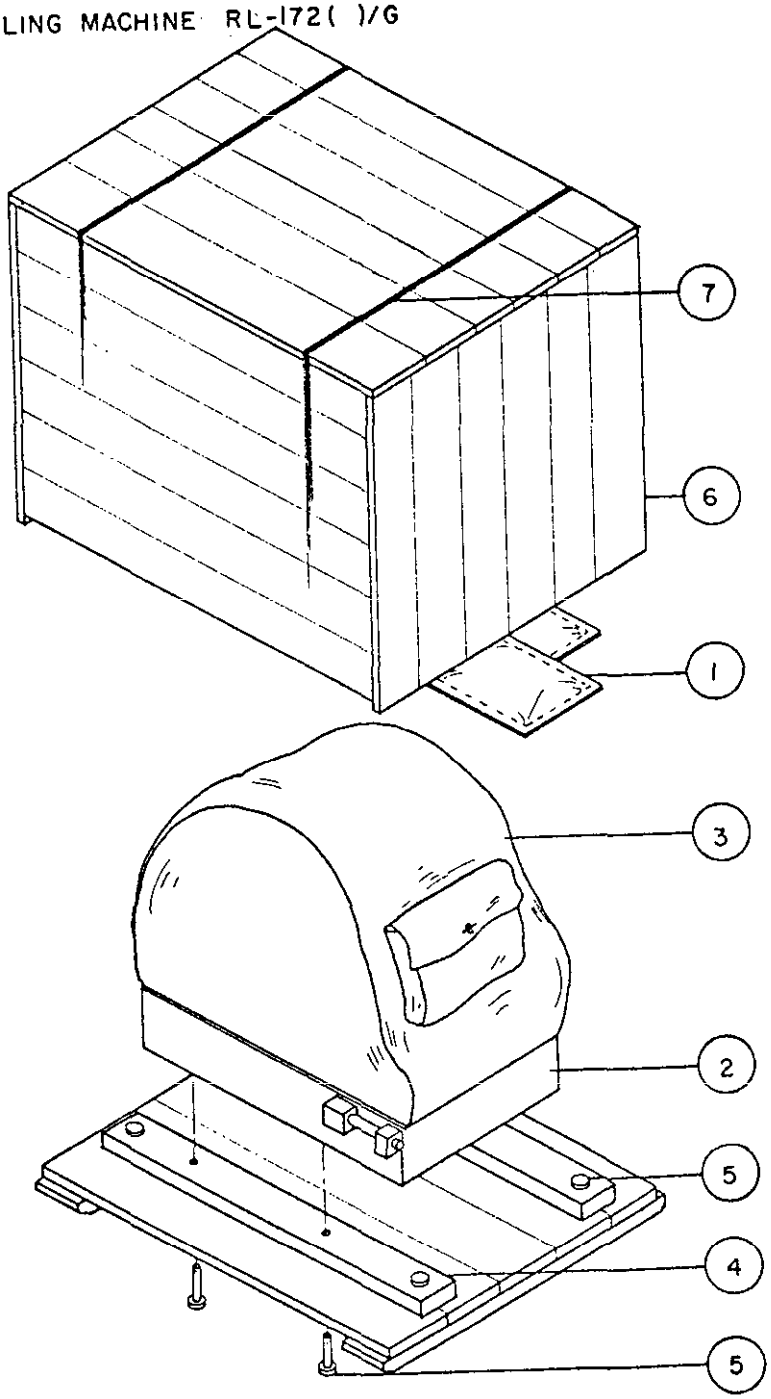


FIGURE 1

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BILL OF MATERIAL
TABLE VI

Ref No	Item Nomenclature	No Rqd	Method MIL-P 116	Size Inches L	W	D	Requirement Spec No.	T	C	V	G	F	S
1	Tech Manual	2	IC-6										
	Bag, barrier	2		11-1/2	9		MIL-B-117	II	D				
2	Reeling Machine RL-172()/G	1		27-1/4	23-1/8	21-1/2	Bare weight 124 1/2 lbs						
	Cable	1	III										
	Cushioning	1		15	9		PPP-P-291	1 (Form over plug)					
	Tape, P.S.	1		12	3		PPP-T-60	1 and tape)					
	Tape, filament	2		14	1/2		PPP-T-97	IV					
		1		30	1/2								
			Secure cable to unit										
	Cable guide	1	III										
	Cushioning	1		22	15		PPP-P-291	I					
	Tape, P.S.	2		6	3		PPP-T-60	1					
		2		17	3								
			Secure cable guide to unit.										
	Pad LC-80	1	III										
	V Belt	1	III										
	Wrench	1	III										
	Cushioning			16	14	1/2	PPP-C-843	II	A				
	Paper, wrapping	1		10	13		UU-P-271			C-1			
	Tape, P.S.	1		26	4		PPP-T-60			1			
			Wrap items in cushioning and paper, seal paper wrapping at edges with tape, and place within the canvas cover pocket.										
3	Cover, canvas	1	Place over unit.										
4	Blocking lumber	2		27-7/8	4	2 nom	MIL-STD-731						
5	Bolts, nuts and washers	4sets		3-1/2	3/8dia		FF-B-571	C2					
			Secure through floorboards and blocking lumber to item.										
		4 sets		4-1/2	3/8dia								
			Secure through skids, floorboards, and ends of blocking lumber. Recess heads.										
6	Box, wood, nailed and lock corner	1		28	24	24-1/2	PPP-B-621	2					7
			Screws shall be used for closure.										
7	Strapping, metal			117	3/8	.020	QQ-S-781	1	B		2		

Lev.A PACKED WEIGHT - 190 lbs
Lev.A PACKED VOLUME 13.15 cu ft

T-Type V-Variety F-Flute
C-Class G-Grade S-Style

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6. NOTES

6.1 Intended use.- Reeling Machine, RL-172( )/G is intended for use by troops to payout field wire from a mobile configuration and pick up field wire from either a mobile or a stationary configuartion.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

(c) Levels of preservation, packaging, packing and marking. (See Section 5).

(d) Marking and shipping of samples.

(e) Place of final inspection.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: RL-172W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.4.3.3)

6.5 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive Quality Assurance Provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN

ARMY EL

Preparing activity

ARMY EL

Project No. 3895-A064

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HEADQUARTERS
U.S.ARMY ELECTRONICS COMMAND
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

To detach this form, cut along this line

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